University of Florida, Department of Mathematics

Algebra First-Year Exam, August 2023, Part 2

Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Suppose R is a nonzero commutative ring with identity, and recall that $x \in R$ is nilpotent if $x^n = 0$ for some n (possibly depending on x).

(i) Show that the set I of nilpotent elements of R is a proper ideal.

(ii) Show that R/I has no nonzero nilpotent elements.

2. Which of the following polynomials are irreducible in the given ring? Explain, and factor the ones that are not irreducible.

(a) $X^4 + 6X^3 - 9X - 21$ in $\mathbb{Q}[X]$.

(b) $X^3 + 2X^2 + X + 1$ in $\mathbb{Z}[X]$.

(c) $X^3 + Y^2 - Y + 1$ in $\mathbb{Z}[X, Y]$.

3. Let R be an integral domain.

(i) Show that any prime element of R is irreducible.

(ii) Show that if R is a PID then any irreducible element is prime.

(iii) Show that $\mathbb{Z}[\sqrt{-5}]$ is an integral domain, 2 is irreducible, and 2 is not prime.

4. Assume that F is a field. Find the minimal prime ideals of the ring

$$R = F[x_1, x_2, x_3, x_4, x_5, x_6]/(x_1x_2, x_3x_4, x_5x_6).$$

(A minimal prime ideal is a prime ideal that does not properly contain any other prime ideal.)

5. Suppose L/K and E/L are algebraic extensions of fields. Show that E/K is algebraic.

6. This problem has two parts.

(a) Find representatives in rational canonical form for all conjugacy classes in $GL_6(\mathbb{Q})$ with characteristic polynomial $(X^3 - 1)^2$.

(b) For each representative, give the Jordan normal form in $GL_6(\mathbb{C})$.